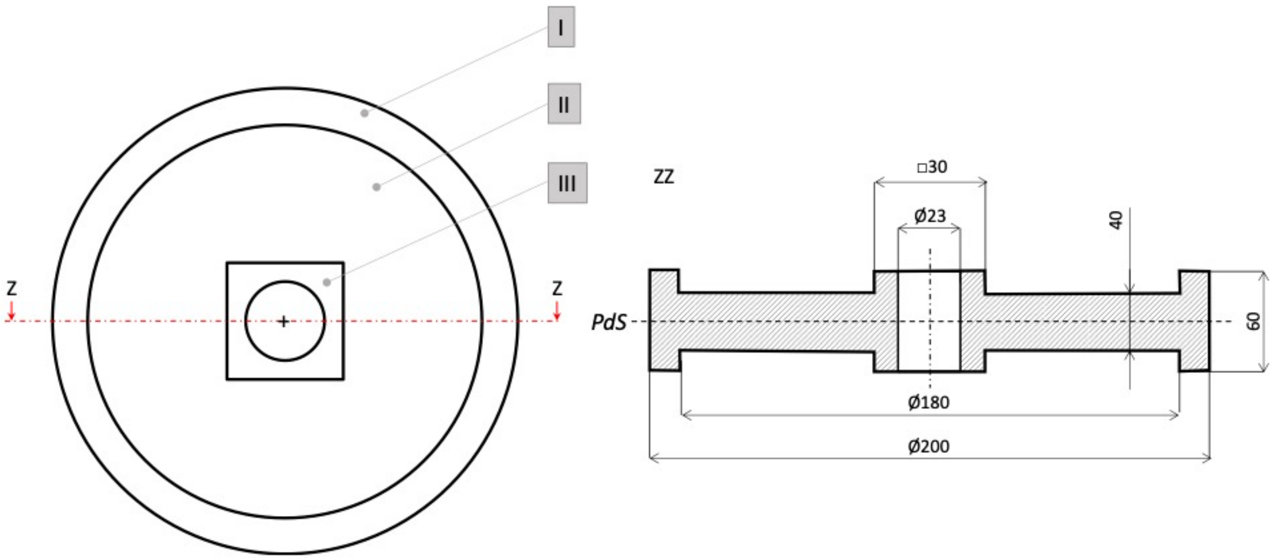
CAM – EXAM 12/07/2021

WRITTEN TEST AY 2020/21

SOLUTIONS TO NUMERICAL PROBLEMS

METAL CASTING



A steel lift pulley is obtained by sand casting using a middle gating system. The top view and the cross-section of the main cavity in the mold are shown in the figure, where PdS is indicating the parting line. In the figure the decomposition into elementary geometries is also indicated (draft angles and fillet radii are neglected).

1) Study the directional solidification of the casting.

2) Considering a volume of the entire cavity equal to 1,25 · 10^6 mm^3, a total volume of open risers equal to 400 cm^3, an initial pouring height equal to 100 mm, a friction factor c = 0,6, and a gating system ratio $As{:}Ar{:}Ag$ = 4:3:2, design the gating system to ensure a mold filling time equal to 10 s.

Solution

1) Study the directional solidification of the casting.

Element I:

$$V_I = \frac{\pi}{4}(200^2 - 180^2) \cdot 60 = 358.142 \text{ mm}^3$$

$$A_I = 2 \cdot \frac{\pi}{4}(200^2 - 180^2) + \pi \cdot 200 \cdot 60 + \pi \cdot 180 \cdot (60 - 40) = 60.947 \text{ mm}^2$$

$$M_I = \frac{V_I}{A_I} = \frac{358.142}{60.947} = 5{,}9 \text{ mm}$$

Element II:

$$V_{II} = \frac{\pi}{4}180^2 \cdot 40 - 30^2 \cdot 40 = \left(\frac{\pi}{4}180^2 - 30^2\right) \cdot 40 = 981.876 \text{ mm}^3$$

$$A_{II} = 2 \cdot \left(\frac{\pi}{4}180^2 - 30^2\right) = 49.094 \text{ mm}^2$$

$$M_{II} = \frac{V_{II}}{A_{II}} = \frac{981.876}{49.094} = 20 \text{ mm}$$

Element III:

$$V_{III} = 30^2 \cdot 60 - \frac{\pi}{4}23^2 \cdot 60 = \left(30^2 - \frac{\pi}{4}23^2\right) \cdot 60 = 29.071 \text{ mm}^3$$

$$A_{III} = 2 \cdot \left(30^2 - \frac{\pi}{4} \cdot 23^2\right) + 4 \cdot 30 \cdot 20 + \pi \cdot 23 \cdot 60 = 7.704 \text{ mm}^2$$

$$M_{III} = \frac{V_{III}}{A_{III}} = \frac{29.071}{7.704} = 3{,}8\ \mathrm{mm}$$

Directional solidification: The first element that will solidify is Element III, then Element I and at last Element II. Being M_{II} too high with respect both M_I and M_{III} even if the directional solidification will be guaranteed, fracture may occur during the cooling at solid state.

2) *Considering a volume of the entire cavity equal to 1,25 · 10⁶ mm³, a total volume of open risers equal to 400 cm³, an initial pouring height equal to 100 mm, a friction factor* $c = 0{,}6$, *and a gating system ratio* $As\!:\!Ar\!:\!Ag = 4\!:\!3\!:\!2$, *design the gating system to ensure a mold filling time equal to 10 s*.

Being the mold filling time equal to $T_{MF} = \frac{V_{total}}{A_{bn} \cdot v}$, we know that the minimal bottleneck cross section (A_{bn}), that is the area of the gates in this situation, will be $A_{bn} = A_g = \frac{V_{total}}{T_{MF} \cdot v}$.

Therefore:

$$V_{total} = V_{casting} + V_{risers} = 1.250.000 + 400.000 = 1.650.000\ \mathrm{mm}^3$$

The pouring velocity in the bottleneck cross section will be $v = c \cdot \sqrt{2 \cdot g \cdot H_m}$, where

$$H_m = \frac{1}{\left(\frac{r'}{\sqrt{h_i}} + \frac{r''}{\sqrt{h_m}}\right)^2}$$

Being $h_i = 100\ \mathrm{mm}$ and $h_f = 0\ \mathrm{mm}$ (we are considering open risers), we get:

$$h_m = \left(\frac{\sqrt{h_i} + \sqrt{h_f}}{2}\right)^2 = \left(\frac{\sqrt{100} + \sqrt{0}}{2}\right)^2 = 25\ \mathrm{mm}$$

$$r' = \frac{V_{casting}/2}{V_{total}} = \frac{1.250.000/2}{1.650.000} = 0{,}38$$

$$r'' = 1 - r' = 1 - 0{,}38 = 0{,}62$$

$$H_m = \frac{1}{\left(\frac{r'}{\sqrt{h_i}} + \frac{r''}{\sqrt{h_m}}\right)^2} = \frac{1}{\left(\frac{0{,}38}{\sqrt{100}} + \frac{0{,}62}{\sqrt{25}}\right)^2} = 38{,}1\ \mathrm{mm}$$

$$v = c \cdot \sqrt{2 \cdot g \cdot H_m} = 0{,}6 \cdot \sqrt{2 \cdot 9{,}81 \cdot \frac{38{,}1}{1000}} = 0{,}519\ \frac{\mathrm{m}}{\mathrm{s}} < 1\frac{\mathrm{m}}{\mathrm{s}}$$

Therefore:

$$A_g = \frac{V_{total}}{T_{MF} \cdot v} = \frac{1.650.000}{10 \cdot 0{,}519 \cdot 1000} = 318\ \mathrm{mm}^2$$

$$A_r = \frac{3}{2}A_g = \frac{3}{2} \cdot 318 = 477\ \mathrm{mm}^2$$

$$A_s = \frac{4}{2}A_g = 2 \cdot 318 = 636\ \mathrm{mm}^2$$

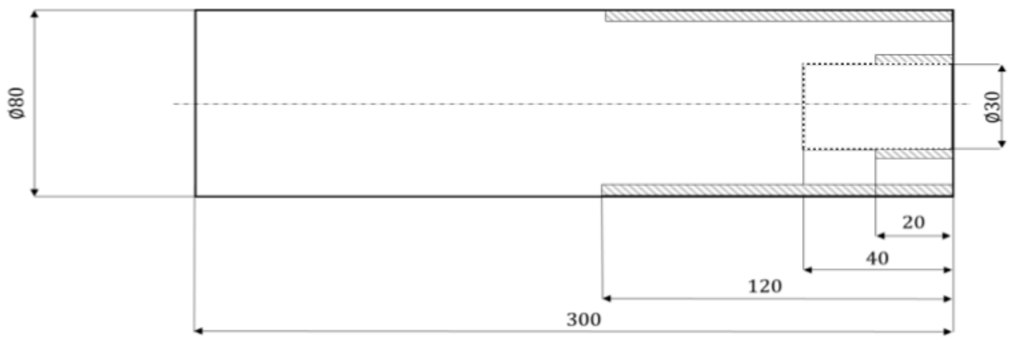
TURNING

The mechanical component in the figure must be machined on a lathe. The machining operations are an external cylindrical turning of a length equal to 120 mm and an internal cylindrical turning of a length equal to 20 mm. A metal bar of 80 mm diameter is mounted on a self-centering 3-jaw chuck and machined according to the following parameters (the two machining operations are carried out with tools with identical characteristics):

- External cylindrical turning: depth of cut = 1 mm, feed 0,4 mm/rev.
- Internal cylindrical turning: depth of cut = 0,5 mm, feed 0,1 mm/rev.
- Other data: Cutting pressure: kc0,4 = 2600 MPa, Kronenberg coefficient x = 0,29, Main entering angle = 95°, Secondary entering angle = 10°, Tool nose radius = 0,4 mm, Nominal lathe power = 5 kW, Lathe efficiency = 90%.

1) Determine the cutting force, the cutting power and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.

2) Determine the average roughness on the machined surface of the hole.



Solution

1) *Determine the cutting force, the cutting power and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.*

Using the cutting pressure method, the cutting force is equal to $F_C = k_c \cdot A_D$, where

$$k_c = k_{c,0,4}\left(\frac{0{,}4}{f \cdot \sin k_r}\right)^x = 2600 \cdot \left(\frac{0{,}4}{0{,}4 \cdot \sin 95°}\right)^{0{,}29} = 2.603 \text{ MPa}$$

$$A_D = f \cdot a_p = 0{,}4 \cdot 1 = 0{,}4 \text{ mm}^2$$

Therefore

$$F_C = 2.603 \cdot 0{,}4 = 1041 \text{ N}$$

The cutting power is

$$P_C = v_c \cdot F_c = \frac{250 \cdot 1041}{60} = 4.338 \text{ W}$$

The constraint on the cutting power is

$$P_{limit} = P_g \cdot E = 5 \cdot 0{,}9 = 4{,}5 \text{ kW}$$

Being $P_C \leq P_{limit}$, the external cylindrical turning is feasible.

The cutting time is equal to $t_c = \frac{l_c}{v_f}$, where, from the figure,

the cutting length is

$$l_c = 120 + 1 = 121 \text{ mm}$$

the final diameter is

$$D_f = D_0 - 2 \cdot a_p = 80 - 2 \cdot 1 = 78 \text{ mm}$$

the rotational speed is

$$n = \frac{1000 \cdot v_c}{\pi \cdot D_f} = \frac{1000 \cdot 250}{\pi \cdot 78} = 1020 \text{ RPM}$$

and the feed velocity is

$$v_f = f \cdot n = 0{,}4 \cdot 1020 = 408 \text{ mm/min.}$$

Therefore,

$$t_c = \frac{l_c}{v_f} = \frac{121}{408} \cdot 60 = 17{,}8 \text{ s}$$

2) Determine the average roughness on the machined surface of the hole.

Using the Schmalz equation, we get

$$R_a = \frac{1}{32} \cdot \frac{f^2}{r} \cdot 10^3 = \frac{1}{32} \cdot \frac{0{,}1^2}{0{,}4} \cdot 10^3 = 0{,}8 \text{ µm}$$

To apply this equation, the following constraints must be verified:

$$f = 0{,}1 < 2 \cdot r \cdot \sin(k_r) = 2 \cdot 0{,}4 \cdot \sin(95°) = 0{,}8 \text{ mm/rev.}$$

$$f = 0{,}1 < 2 \cdot r \cdot \sin(k_r') = 2 \cdot 0{,}4 \cdot \sin(10°) = 0{,}14 \text{ mm/rev.}$$

The maximum feed limit is $f_{limit} = 0{,}14$ mm/rev, so the equation can be applied.

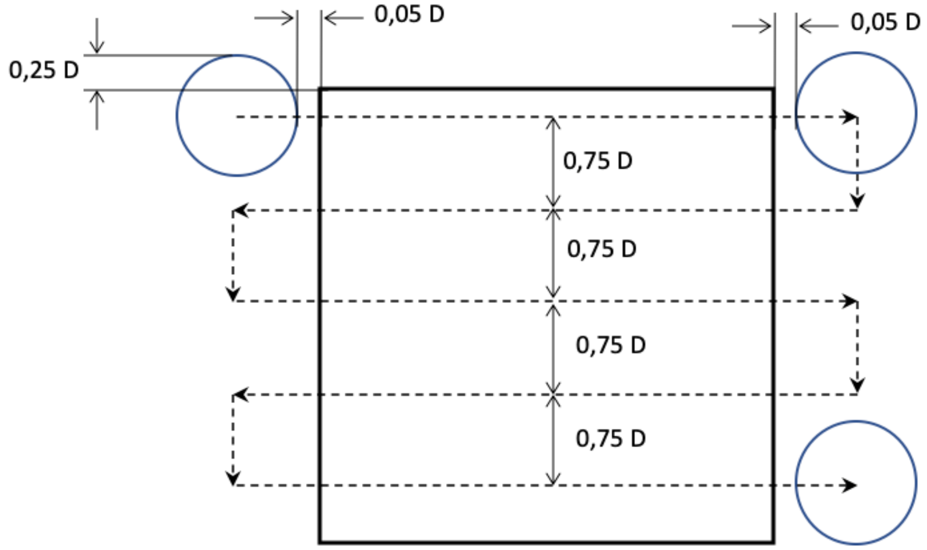
MILLING

A steel square planar surface with a length equal to 300 mm must be flattened. A roughing operation will remove a machining allowance equal to 4 mm. The face mill will be positioned initially so that 25% of the tool diameter will exit from one side of the plane width and the square plane is machined according to the path in the figure.

Determine:

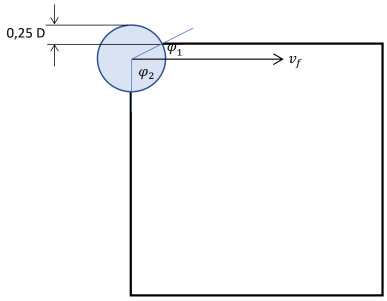
1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 15 kW and an efficiency of 80%.
2) The machining time needed to carry out the operation.

Data: mill diameter D = 80 mm; number of mill cutting edges Z = 8; main entering angle equal to 90°; Kronenberg parameters k_{cs} = 1600 MPa, x = 0,197; feed per tooth f_z = 0,15 mm/rev; cutting speed v_c = 250 m/min.



Solution

1) *The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 15 kW and an efficiency of 80%.*



To answer, it is necessary to understand what the actual situation in this face milling operation is. First, we need to compute the engagement angle and then the number of simultaneously working teeth.

Referring to the figure, the overall engagement angle is the sum of the following two angles:

$$\varphi_1 = \arcsin\left(\frac{D/2 - 0{,}25 \cdot D}{D/2}\right) = \arcsin\left(\frac{80/2 - 20}{80/2}\right) = \arcsin\left(\frac{1}{2}\right) = 30°$$

$$\varphi_2 = 90°$$

$$\varphi = \varphi_1 + \varphi_2 = 30° + 90° = 120° = 2{,}1 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{360}{8} = 45° = 0{,}785 \text{ rad}$$

$$z = \frac{\varphi}{\varphi_0} = \frac{120°}{45°} = 2{,}67$$

Therefore, we must consider the average approach, where $a_e = 0{,}75 \cdot D = 60$ mm. Therefore:

$$h_{D,avg} = \frac{f_Z}{\varphi} \cdot \frac{2 \cdot a_e}{D} = \frac{0{,}15}{2{,}1} \cdot \frac{2 \cdot 60}{80} = 0{,}107 \text{ mm}$$

$$A_{D,avg} = h_{D,avg} \cdot a_p = 0{,}107 \cdot 4 = 0{,}428 \text{ mm}^2$$

$$k_{c,avg} = k_{cs} h_{D,avg}^{-x} = 1.600 \cdot (0{,}107)^{-0{,}197} = 2.485 \text{ MPa}$$

$$F_{c,avg} = k_{c,avg} A_{D,avg} = 2.485 \cdot 0{,}428 = 1.064 \text{ N}$$

$$T_c = z \cdot F_{c,avg} \frac{D/2}{1000} = 2{,}67 \cdot 1.064 \cdot \frac{80/2}{1000} = 113{,}64 \text{ Nm}$$

Considering a cutting speed $v_c = 250 \ \frac{\text{m}}{\text{min}}$, we get

$$\omega = \frac{1000}{60} \cdot \frac{v_c}{D/2} = \frac{1000}{60} \cdot \frac{250}{80/2} = 104{,}2 \ \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 113{,}64 \cdot 104{,}2 = 11.841 \text{ W} = 11{,}8 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0{,}8 \cdot 15 = 12 \text{ kW}$$

Therefore, being $P_c = 11{,}8 \text{ kW} \leq P_{c,limit} = 12 \text{ kW}$, the face milling operation is feasible.

2) *The cutting time needed to carry out the operation.*

The machining time is equal to $t_c = \frac{l_m}{v_f}$. Referring to the figure, we have

$$l_m = \left(1{,}1 \cdot D + l_{workpiece}\right) \cdot 5 + 0{,}75 \cdot D \cdot 4$$

$$l_m = (1{,}1 \cdot 80 + 300) \cdot 5 + 0{,}75 \cdot 80 \cdot 4 = 388 \cdot 5 + 240 = 2.180 \text{ mm}$$

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 250}{\pi \cdot 80} = 995 \text{ RPM}$$

$$v_f = Z \cdot f_Z \cdot n = 8 \cdot 0{,}15 \cdot 995 = 1.194 \ \frac{\text{mm}}{\text{min}}$$

Therefore:

$$t_m = \frac{2.180}{1.194} = 1{,}83 \text{ min} = 110 \text{ s}$$